

Mode of Examination: Online
M.Sc. Semester – I Examination, 2022

2022

Subject: Computer Science

Paper Code & Name: CSMC104 (Object Oriented Analysis and Design)

Full Marks: 70

Date: 09.03.2022

Time and Duration: 4:30 PM-7:30 PM (3:00 Hours)

Please note the following instructions carefully:

- **Promise not to commit any academic dishonesty.**
- **Marks will be deducted if the same/similar answers are found in different answer scripts.**
- **Candidates are required to answer in their own words as far as applicable.**
- **The figures in the margin indicate full marks.**
- **Each page of the answer-scripts should have your University Roll # on the right-top corner.**
- **The name of the scanned copy of the answer script will be of the following format:**
CSMC104-OOAD-University-Roll-Number.pdf
(Example: CSMC104-OOAD-C91-CSC-2110xy.pdf)
- **The subject of the mail should be the file name only.**
- **The scanned answer script is to be sent to cucse2020@gmail.com**
- **The file should have the top page (Page #1) as an index page, mentioning all page numbers against the answer of each question number.**
- **The answer script may not be accepted after the scheduled time.**

Answer *Question Nos. 1 and 2*, and *any four* from the rest.

1. Answer any five questions. **2x5=10**
 - a. What is the use of Demon Thread? Give an example.
 - b. Identify the type of cohesion in 'updating a record in database and send it to printer'.
 - c. What do you mean by incomplete generalization? Give an example.
 - d. A folder may contain zero or more files – represent this using appropriate UML diagram.
 - e. What is the basis for classifying design patterns? Tabulate at least two design patterns of each category.
 - f. Name at least one class in JDK that uses factory design pattern and at least one class in JDK that uses observer pattern.

2. Answer any five questions. **5x4=20**
 - a. Identify major differences among Aggregation, Association and Composition with suitable example.
 - b. With a suitable example explain how and when to use singleton pattern.
 - c. Compare and contrast static binding and dynamic binding with suitable example.
 - d. Critically comment on the statement – “Java is Platform independent and Secure”.
 - e. Discuss the in-memory structure of JDK. Show the memory content and references when you create an object using its default constructor.
 - f. A small library database system will be used by CSE, IT and ECE departments. The library contains books and other media such as videos and software. A secretary will operate the system and is responsible for checking out books and other media to students and faculty members. Identify the objects and relation among the objects.

3. (a) Differentiate between functional programming and object-oriented programming.
 (b) What do you mean by compile time polymorphism and runtime polymorphism? How are they implemented in Java? Explain your answer with a suitable example.
 (c) Explain the phrase 'Single interface multiple implementation' in the context of designing API (Application Programming Interface). **[3+(2+3)+2]**

4. (a) Define inheritance. List various ways in which inheritance can be implemented in Java.
 (b) Make a class Student. The Student class has data members namely roll, name and branch. Create a class called Subject with data members code, name and credit point. Create a class called Result to store the marks of 6 different subjects of a student. The Result class can be used to find total marks, average marks and total credit points earned by a student. Write a program in your preferred language for the above problem modelling the relationships among the classes Student, Subject and Result. **[(2+2)+6]**

5. Use appropriate design patterns to design (no need to implement) an Employee Management System (EMS) of an organization having hierarchical employee structure. Define and implement an API so that a client can send the following queries to your EMS using your API:
 - i. Find total employees and total salary of all employees of the organization.
 - ii. Find total employees and total salary of all employees in a department of the organization.
 - iii. Joining and Resignation of employees. **10**

6. (a) Why do we use Factory design pattern? Differentiate between Factory and Factory Method design pattern.
 (b) Consider building a Logger Framework where the log messages maybe written into the log file (represented by FileLogger class) or displayed in the output console (represented by ConsoleLogger class) or written to an error console (represented by ErrorLogger). Design a software using Factory Method design pattern so that a client can log a message by simply invoking LogFatory.log(type, msg) where 'type' indicates type of logger to be used to log the message 'msg'. **[(1+2)+7]**

7. Discuss the problem statement, solution, consequences and trade-offs of Observer design pattern with a suitable example. **10**